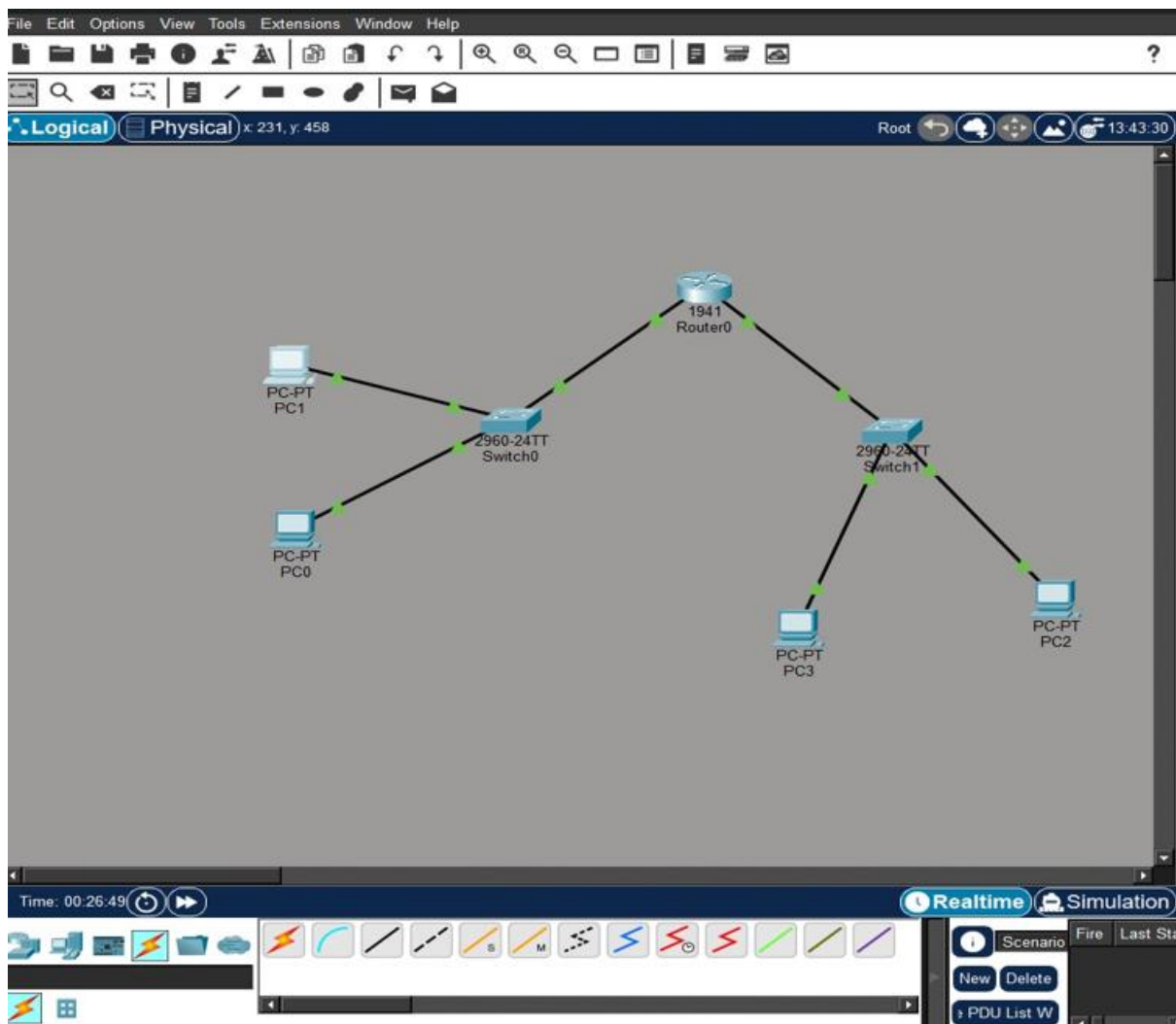


EXPERIMENT-10

Internetworking with routers in CISCO PACKET TRACER simulator.

a) Design and configure a simple internetwork using a router:

1. Design different networks (with 3 to 4 hosts) and connect via Router.
2. Allot static ip address to machines and router interfaces.
3. Perform simulation and trace how routing is done in packet transmission.



EXPERIMENT-10

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
```

Copy

Paste

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

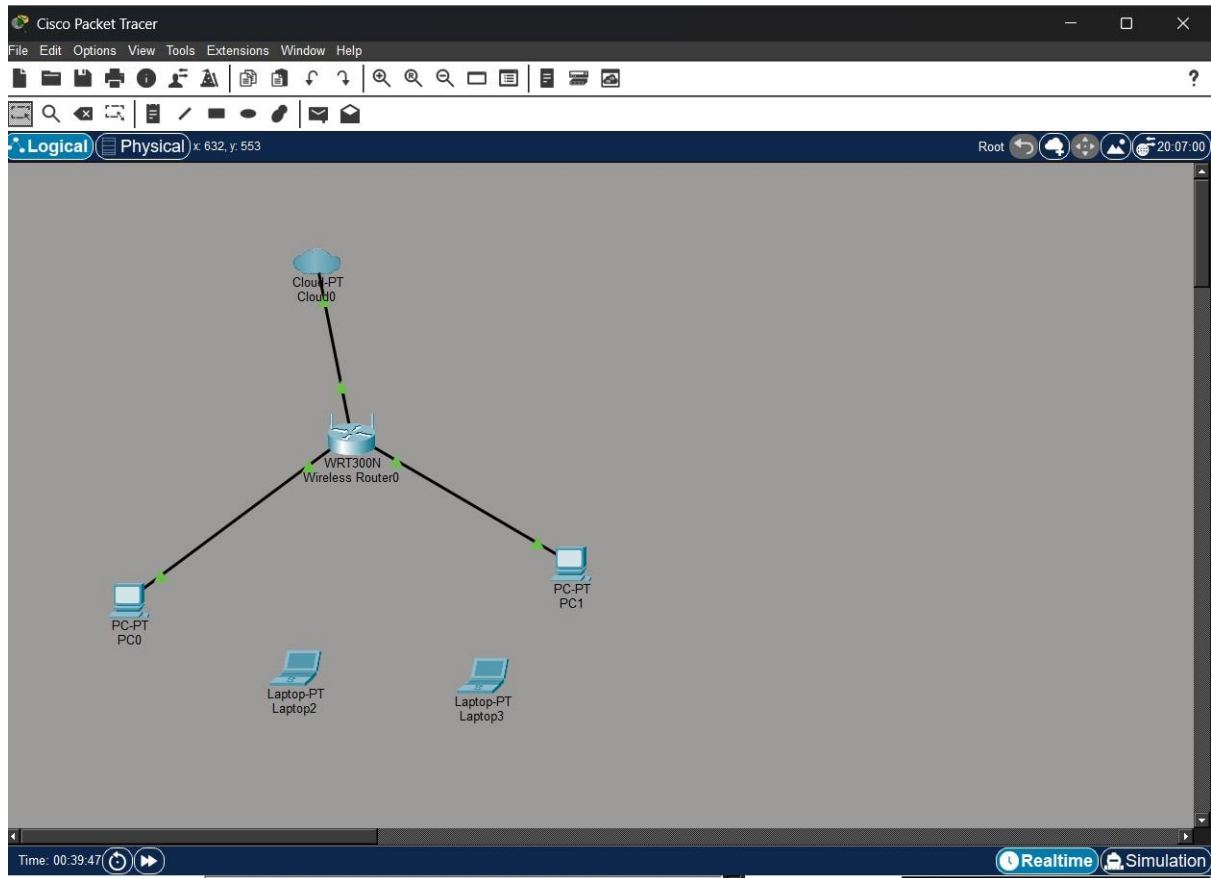
Request timed out.
Reply from 192.168.2.3: bytes=32 time=1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

EXPERIMENT-10

b) Design and configure an internetwork using wireless router DHCP server and internet cloud.



EXPERIMENT-10

c) Design and configure an inter-network in your lab using switch, router and Ethernet cables.

```
Router0
Physical Config CLI Attributes

170 West Tasman Drive
San Jose, California 95134-1706

Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2015 by Cisco Systems, Inc.
Compiled Thu 9-Jun-12 15:41 by pt team
Image text-base: 0x2100F910, data-base: 0x24729040

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wd/export_crypto_tool/faq.cgi.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco C1900-1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTK152400P3
2 Gigabit Ethernet interfaces
DRAM configuration is 64 Bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
4096K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

ALINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
ALINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
NS13-S-CONFIG_1_1 Configured from console by console
```

```
PC3
Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=5ms TTL=128
Reply from 192.168.2.3: bytes=32 time=5ms TTL=128
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 3ms

C:\>
```